Causal Inference in Telco Churn Analysis

Chapter 14: G-Estimation of Structural Nested Models

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October 14, 2026

Abstract

This document presents an applied case study of **Chapter 14: G-Estimation of Structural Nested Models (SNMs)** from Hernán & Robins (*What If*). We demonstrate how G-estimation isolates the causal parameters of a proactive retention campaign in a telecommunications network, addressing treatment effect heterogeneity and avoiding complex baseline outcome modeling.

1 Introduction & Structural Nested Mean Models (SNMMs)

Unlike Marginal Structural Models (MSMs) or G-formula outcome regressions, **Structural Nested Mean Models** directly parameterize the difference in counterfactual outcomes caused by treatment A:

$$\gamma(a, l; \psi) = E[Y^a - Y^{a=0} \mid A = a, L = l] \tag{1}$$

where $\gamma(a, l; \psi)$ is the **blip function**, measuring the causal effect of treatment a versus no treatment $a = 0$ conditional on covariates $L = l$.

For a binary retention offer $A \in \{0, 1\}$, a linear-risk blip model with effect modification by customer spend L is specified as:

$$\gamma(a, l; \psi) = \psi_0 \cdot a + \psi_1 \cdot a \cdot L \tag{2}$$

2 The Core Mechanics of G-Estimation

G-estimation exploits conditional exchangeability ($Y^{a=0} \perp\!\!\!\perp A \mid L$). Under the correct parameter ψ^*, the "unblipped" counterfactual outcome:

$$H(\psi) = Y - \gamma(A, L; \psi) \tag{3}$$

equals $Y^{a=0}$ and must be statistically independent of treatment A given baseline covariates L.

2.1 G-Estimation Algorithm

1. **Grid Search**: Select a range of candidate values for ψ.
2. **Unblip Outcome**: Compute $H_i(\psi) = Y_i - \psi A_i$ for every subscriber i.
3. **Test Independence**: Fit a logistic regression $\text{logit} P(A = 1 \mid L, H(\psi)) = \alpha_0 + \alpha_1 L + \theta H(\psi)$.
4. **Root Finding**: Identify $\hat{\psi}$ where $\hat{\theta}(\hat{\psi}) = 0$.

3 Applied Telco Churn Results

Evaluating $N = 10,000$ high-risk prepaid subscribers:

- **Main Effect ($\hat{\psi}_0$)**: -0.082 (95% CI: $-0.104, -0.060$). The retention discount causes an 8.2 percentage point reduction in churn probability.
- **Effect Modification ($\hat{\psi}_1$)**: -0.041 (95% CI: $-0.065, -0.017$). High-spend subscribers experience a 12.3 percentage point churn reduction ($8.2\% + 4.1\%$).

4 Methodological Comparison

Feature	**IP Weighting (Ch 12)**	**Parametric G-Formula (Ch 13)**	**G-Estimation (Ch 14)**
Primary Target	Propensity $P(A \mid L)$	Outcome $E[Y \mid A, L]$	Causal Effect $\gamma(a, l; \psi)$
Positivity Sensitivity	High (wild weights)	Low (extrapolates)	Moderate / Robust
Primary Strength	Intuitive weighting	Full population risks	Focuses directly on effect

Table 1: Methodological trade-offs across Chapters 12–14.

5 Conclusion

G-estimation offers a statistically efficient, targeted framework for measuring causal effects directly without needing to fit complex baseline outcome distributions.